

# Week 6: Lab

## Sequence alignment and phylogenetics

2025-11-17

### Load packages

```
pkgs <- c("ggplot2","devtools","adegenet","phangorn", "ape", "ade4",
          "Biostrings", "DECIPHER", "ggtree", "pwalgn")

for (i in seq_along(pkgs)){
  pkg <- pkgs[[i]]
  tryCatch(if(pkg%in%rownames(installed.packages())){
    cat("Available:", pkg, "\n")
    library(pkg, character.only = TRUE)
    cat("Package loaded:", pkg, "\n")
  }
  else if(!(pkg%in%rownames(installed.packages()))){
    install.packages("BiocManager")
    BiocManager::install(pkg)
    library(pkg, character.only = TRUE)
    cat("Package loaded:", pkg, "\n")
  }
  else{
    cat("Not available:",pkg,">> now installing...", "\n")
    install.packages(pkg, repos = "https://cloud.r-project.org")
    cat("Now available:", pkg, "\n")
    library(pkg, character.only = TRUE)
    cat("Package loaded:", pkg, "\n")
  }
}
```

```
#####
## After running the above, if you cannot find any of the above packages in R,
## or it's giving you an error, then, run them individual as below
#####

#####
## For "ggplot2", "devtools", "adegenet", "phangorn", "ape", "ade4", "BiocManager"
#####
install.packages("ggplot2") # do that for all
library("ggplot2") #load it

#####
## For "Biostrings", "DECIPHER", "ggtree", "pwalign"
#####
BiocManager::install("Biostrings") # do that for all
library(Biostrings) #load it
```

## 1) Sequence Alignment

### a) Pairwise alignment

```
## These are the sequences we are working with for pairwise alignment

## Query sequence
Seqq <- "ATTTGAATGGATGTCAATCCGACTCTACTGTTCTTAAAGGTTCCAGCGCAAAATGCCATAAGCACCACATT
CCCTTATACTGGAGATCCTCCATACAGCCATGGAACAGGAACAGGGTACACCATGGACACAGTCAACAGAACACACCAAT
ATTCAGAGAAGGGGAAGTGGACGACAAATACAGAACTGGGGCACCCCACTCAACCCAATTGATGGACCACTACCTGAG
GATAATGAGCCAAGTGGATATGCACAAACAGACTGTGTCTGGAGGCTATGGCCTTCCTTGAAGAATCCCACCCAGGTAT
CTTTGAGAACTCATGCCTTGAAACAATGGAAGTCGTTCAACAAACAAGGTGGACAACTAACCCAAGGTCGCCAGACTT
ATGATTGGACATTAAACAGAAATCAACCGGCAGCAACTGCATTAGCCAACACCATAGAAGTTTTTAGATCGAATGGACTA
ACAGCTAATGAATCGGGAA" #GenBank: CY002062.1

## Query sequence 2
Seqq2 <- "AGCAAAAGCAGGGGATAATTCTATTAACCATGAAGACTATCATTGCTTTGAGCTACATTCTATGTCTGGT
TTTCGCTCAAAAACCTTCCCGGAAATGACAACAGCAGCGCAACGCTGTGCCTTGGGCACCATGCAGTACCAAACGGAACGA
TAGTGAAAAACAATCACGAATGACCAATTGAAGTCACTAATGCTACTGAACTGGTTCAGAGTTCCTCAACAGGTGGAATA
TGCGACAGTCTCATCAGATCCTTGATGGAGAAAACTGCACACTAATAGATGCTCTATTGGGAGACCCTCAGTGTGATGG
CTTCCAAAATAAGAAATGGGACCTTTTTGTGAACGCAGCAAAGCCTACAGCAACTGTTACCCTTATGATGTGCCGGATT
ATGCCTCCCTTAGGTCACTAGTTGCCTCATCCGGCACACTGGAGTTTAAACAATGAAAGCTTCAATTGGACTGGAGTCACT
CAAAATGGAACAAGCTCTGC" #GenBank: CY002064.1
```

```
## Reference sequence
```

```
Reff <- "ATGAAGACTATCATTGCTTTGAGCTACATTCTATGTCTGGTTTTTCGCTCAAAAACTTCCCGGAAATGACAA  
CAGCACGGCAACGCTGTGCCTTGGGCACCATGCAGTACCAAACGGAACGATAGTGAAAAACAATCACGAATGACCAAATTG  
AAGTCACTAATGCTACTGAACTGGTTCAGAGTTCCTCAACAGGTGGAATATGCGACAGTCCTCATCAGATCCTTGATGGA  
GAAACTGCACACTAATAGATGCTCTATTGGGAGACCTCAGTGTGATGGCTTCCAAAATAAGAAATGGGACCTTTTGT  
TGAACGCAGCAAAGCCTACAGCAACTGTTACCCTTATGATGTGCCGATTATGCCTCCCTTAGGTCAGTGTGCCTCAT  
CCGGCACACTGGAGTTTAAACAATGAAAGCTTCAATTGGACTGGAGTCACTCAAAATGGAACAAGCTCTGCTTGCAAAAGG  
AGATCTAATAACAGTTTCT" #NCBI Reference Sequence: NC_007366.1
```

```
Seq1 <- DNASTring("ATTTGAATGGATGTCAATCCGACTCTACTGTTCTTAAAGGTTCCAGCGCAAAATGCCATAAGCACCACATTCC
```

```
Seq2 <- DNASTring("AGCAAAAGCAGGGGATAATTCTATTAACCATGAAGACTATCATTGCTTTGAGCTACATTCTATGTCTGGTTTT
```

```
Ref <- DNASTring("ATGAAGACTATCATTGCTTTGAGCTACATTCTATGTCTGGTTTTTCGCTCAAAAACTTCCCGGAAATGACAACAG
```

```
pwalgn1 <- pairwiseAlignment(Seq1, Ref)  
pwalgn1
```

Global PairwiseAlignmentsSingleSubject (1 of 1)

pattern: AT-----TTGAATGGATGTCAATCCGAC...AGATCGAATGGACTAACAGCTAATGAATCGGGAA

subject: ATGAAGACTATCATTGCTTTGAGCTACATTCTAT...CTTGCAAAAGGA----GATCTAATAACAGTTTCT

score: -1237.222

## b) Multiple alignment

```
Mseq <- list()
```

```
Mseq[1] <- Seq2
```

```
Mseq[2] <- Seq1
```

```
Mseq[3] <- Ref
```

```
dna0 <- DNASTringSet(Mseq)
```

```
Malign <- AlignSeqs(dna0)
```

Determining distance matrix based on shared 8-mers:

=====

Time difference of 0 secs

Clustering into groups by similarity:

=====

Time difference of 0 secs

Aligning Sequences:

=====

Time difference of 0.02 secs

Iteration 1 of 2:

Determining distance matrix based on alignment:

=====

Time difference of 0 secs

Reclustering into groups by similarity:

=====

Time difference of 0 secs

Realigning Sequences:

=====

Time difference of 0 secs

Alignment converged - skipping remaining iteration.

Malign

DNASTringSet object of length 3:

```
      width seq
[1]   580 AGCAAAAGCAGGGGATAATTCTATTAACCATGAA...CTGC-----
[2]   580 -----...-----
[3]   580 -----ATGAA...CTGCTTGCAAAAGGAGATCTAATAACAGTTTCT
```

## 2) Phylogenetic analysis

```
dna_df <- read.dna("http://adegenet.r-forge.r-project.org/files/usflu.fasta",
                  format = "fasta")
#####
## If dna_df is not loading, download GENET_fasta.RData from the Week 6 folder
## Copy the path from your PC and paste in the load() below. It will be loaded to R
## E.g, load("C:/Users/Kyles/Download/GENET_fasta.RData")
#####

# load("")

## Perform Distance between sequence
dna_dist <- dist.dna(dna_df, model = "TN93")

dna_tree <- as.phylo(hclust(dna_dist, method = "average"))
```

### a) Model selection

```
To_phyD <- phyDat(dna_df)

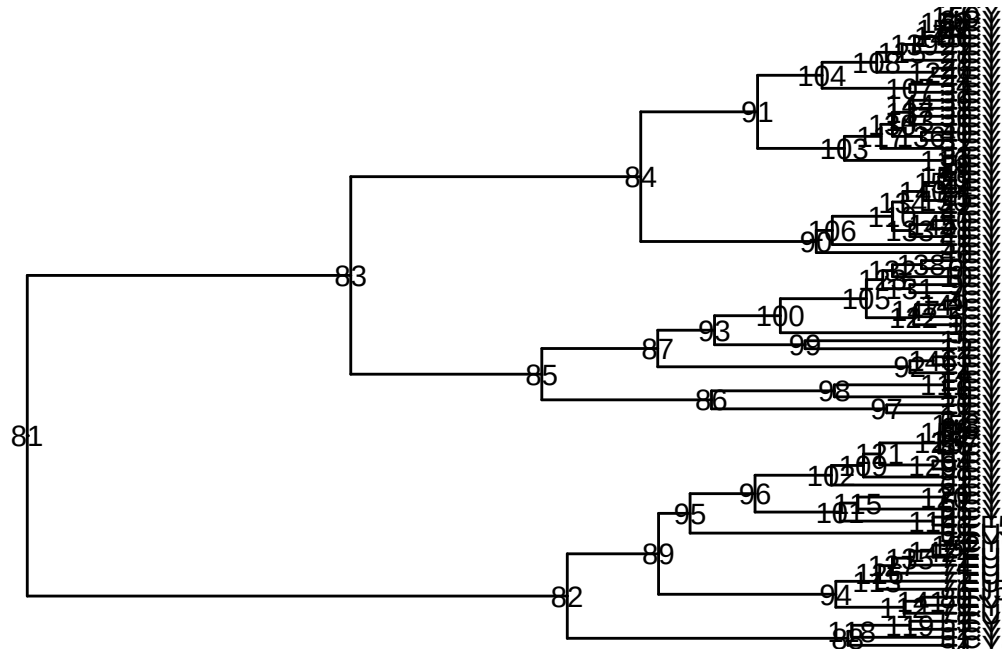
mod.phy <- modelTest(To_phyD, model = c("JC", "F81", "K80", "HKY", "K81",
                                         "SYM", "GTR"))
```

| Model      | df  | logLik    | AIC      | BIC      |
|------------|-----|-----------|----------|----------|
| JC         | 157 | -5634.538 | 11583.08 | 12436.99 |
| JC+I       | 158 | -5575.284 | 11466.57 | 12325.93 |
| JC+G(4)    | 158 | -5573.813 | 11463.63 | 12322.98 |
| JC+G(4)+I  | 159 | -5573.108 | 11464.22 | 12329.01 |
| F81        | 160 | -5592.628 | 11505.26 | 12375.49 |
| F81+I      | 161 | -5533.433 | 11388.87 | 12264.54 |
| F81+G(4)   | 161 | -5532.1   | 11386.2  | 12261.87 |
| F81+G(4)+I | 162 | -5531.471 | 11386.94 | 12268.05 |
| K80        | 158 | -5457.454 | 11230.91 | 12090.27 |
| K80+I      | 159 | -5397.501 | 11113    | 11977.8  |
| K80+G(4)   | 159 | -5395.944 | 11109.89 | 11974.69 |
| K80+G(4)+I | 160 | -5395.17  | 11110.34 | 11980.58 |
| HKY        | 161 | -5415.926 | 11153.85 | 12029.53 |
| HKY+I      | 162 | -5356.431 | 11036.86 | 11917.97 |

|            |     |           |          |          |
|------------|-----|-----------|----------|----------|
| HKY+G(4)   | 162 | -5354.811 | 11033.62 | 11914.74 |
| HKY+G(4)+I | 163 | -5354.065 | 11034.13 | 11920.68 |
| K81        | 159 | -5445.498 | 11209    | 12073.79 |
| K81+I      | 160 | -5385.602 | 11091.2  | 11961.44 |
| K81+G(4)   | 160 | -5384.09  | 11088.18 | 11958.42 |
| K81+G(4)+I | 161 | -5383.344 | 11088.69 | 11964.36 |
| SYM        | 162 | -5436.391 | 11196.78 | 12077.9  |
| SYM+I      | 163 | -5375.934 | 11077.87 | 11964.42 |
| SYM+G(4)   | 163 | -5374.285 | 11074.57 | 11961.12 |
| SYM+G(4)+I | 164 | -5373.444 | 11074.89 | 11966.88 |
| GTR        | 165 | -5395.226 | 11120.45 | 12017.88 |
| GTR+I      | 166 | -5336.013 | 11004.03 | 11906.89 |
| GTR+G(4)   | 166 | -5334.607 | 11001.21 | 11904.08 |
| GTR+G(4)+I | 167 | -5333.943 | 11001.89 | 11910.19 |

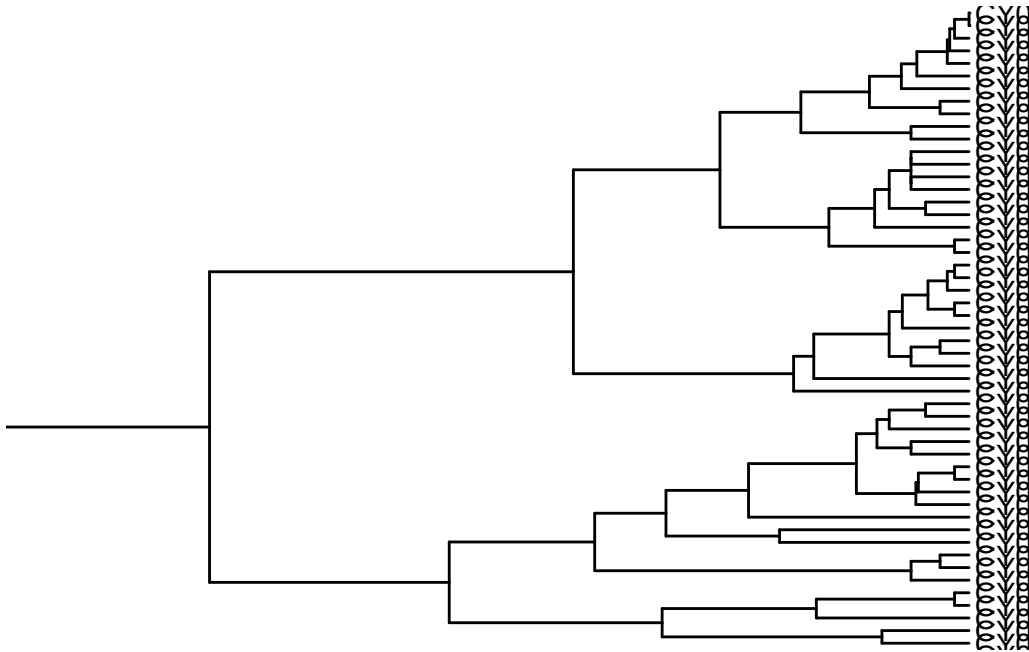
## b) Plot trees

```
## View node numbering
ggtree(dna_tree)+
  geom_tiplab()+
  geom_text2(mapping = aes(label = node))
```



```
## Add tip label layer to the tree
gtree <- ggtree(dna_tree)+
  geom_tiplab()

## View subset of the node
viewClade(gtree, node = 83)
```



```
## Tree layouts

ggtree(dna_tree, layout = "radial")+
  geom_tiplab()
```



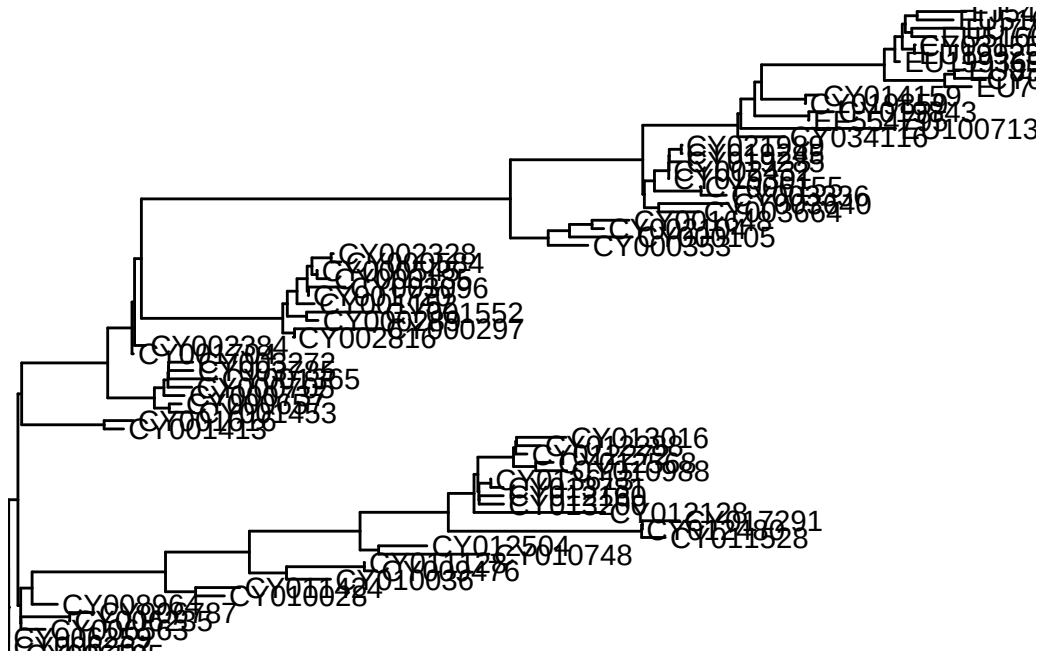
### c) Using Neighbor-joining

```
## Compute tree using neighbor-joining
nj_tree <- nj(dna_dist)

## Convert to phylo object
nj_tree2 <- as.phylo(nj_tree)

## Plot tree
nj_gtree <- ggtree(nj_tree2)+
  geom_tiplab()

## View a subset node
viewClade(nj_gtree, node=83)
```



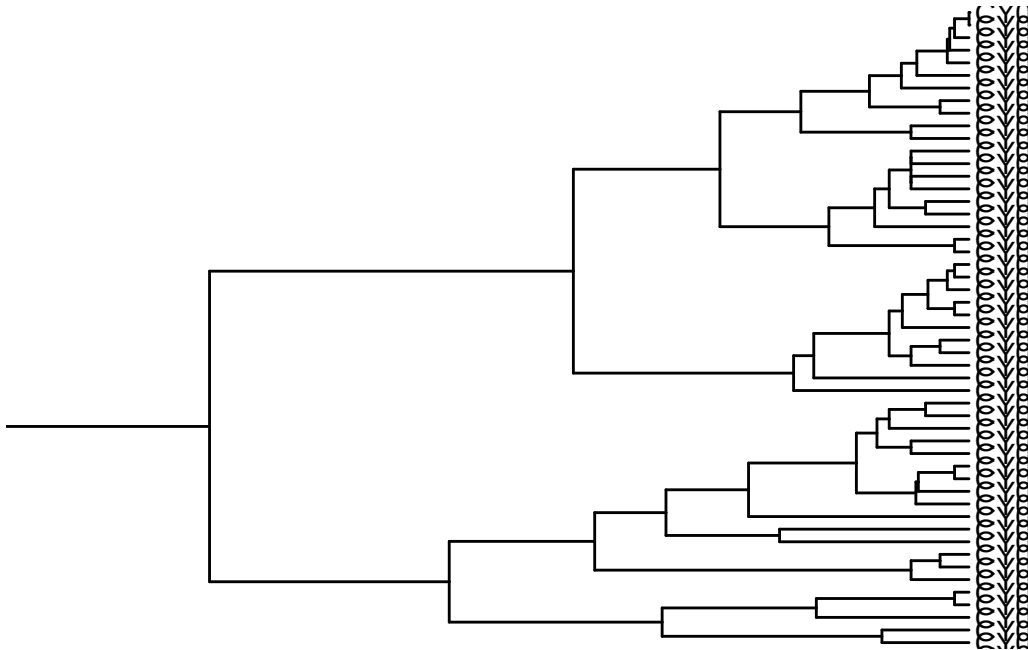
**d) UPGMA**

```
## Compute tree using the unweighted pair group method with arithmetic mean (UPGMA)
upgma_tree <- upgma(dna_dist)
```

```
## Convert to phylo object
upgma_tree2 <- as.phylo(upgma_tree)

## Plot tree
upgma_gtree <- ggtree(upgma_tree2)+
  geom_tiplab()

## View a subset node
viewClade(upgma_gtree, node=83)
```



## Task

- Replace TN93 with the best model (lowest AIC or BIC) in 2a above. That is, choose any of “JC”, “K80”, “F81”, or “K81” model that has the lowest value.
- Compare the tree with 2c and 2d

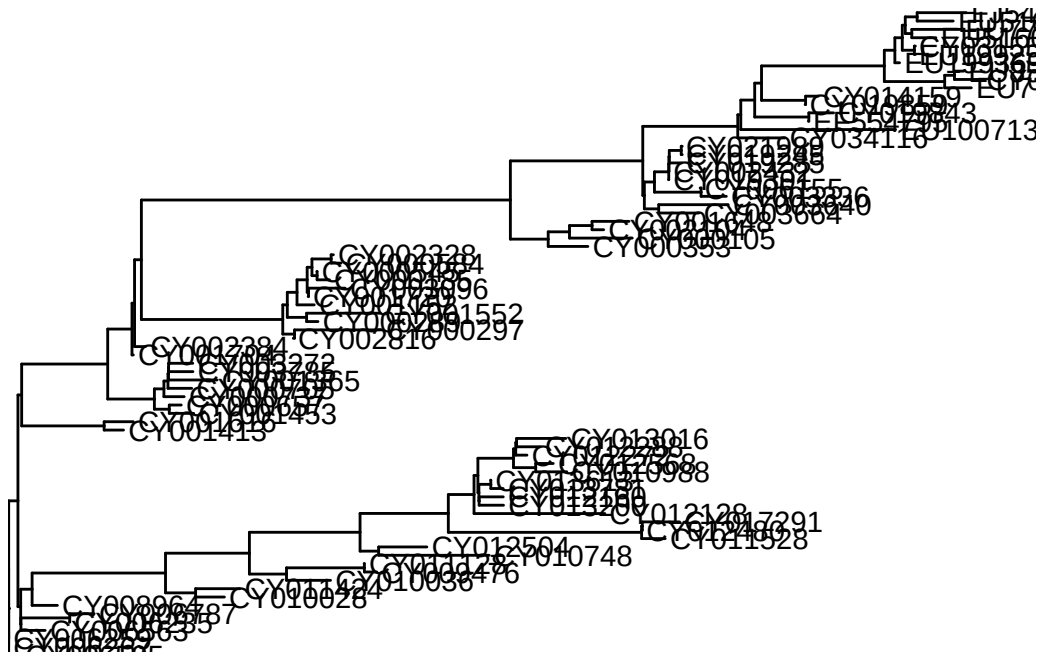
```
## Perform Distance between sequence
dna_mat <- dist.dna(dna_df, model = "K81") #Replace TN93

#####
```

```
nj_tr <- as.phylo(nj(dna_mat))

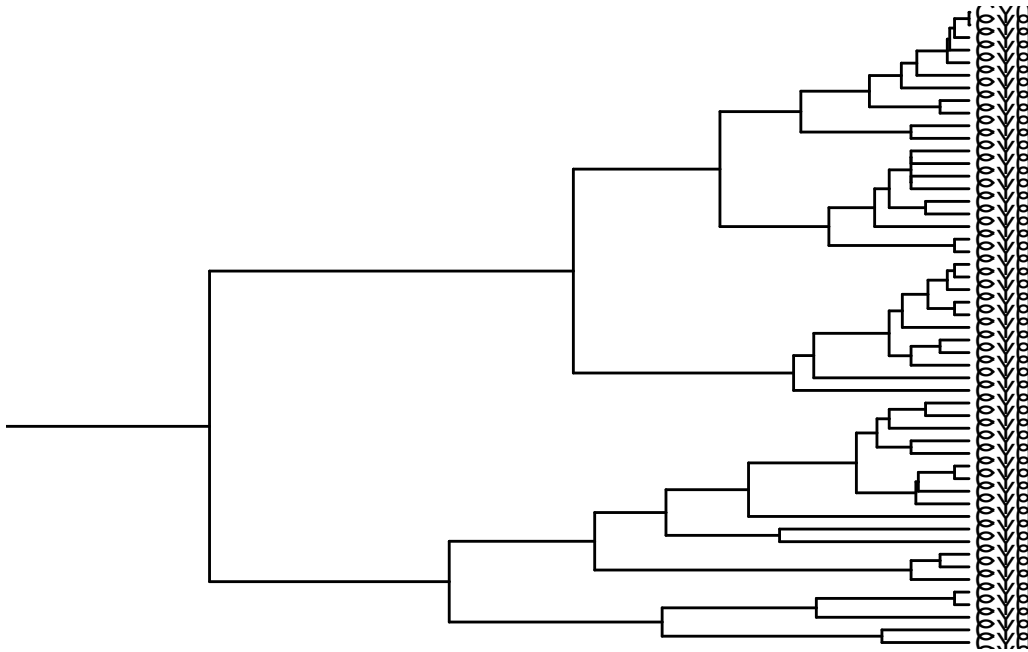
#Plot tree
nj_gtree <- ggtree(nj_tree2)+
  geom_tiplab()

#View a subset node
viewClade(nj_gtree, node=83)
```



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```
## View a subset node  
viewClade(upgma_gtree, node=83)
```



Is there any difference? There's no difference between the task method and 2c&d

Note:

Unlike the UPGMA, the neighbor-joining method does not make all taxa equidistant from the root. This is a crucial difference as UPGMA makes the molecular clock assumption (i.e., uniform evolutionary rates) which NJ doesn't make. This is evident as you'll notice equidistant tips in UPGMA compared to different tip length in NJ both in the task and 2c&d.